

# Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations

Jackie Lok Elizaveta Rebrova

Princeton University



## Subspace-constrained sketch-and-project

Suppose we constrain the dynamics of the *sketch-and-project method* for solving  $\mathbf{Ax} = \mathbf{b}$ ,  $\mathbf{A} \in \mathbb{R}^{m \times n}$ , within the (affine) subspace

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{QA}\mathbf{x} = \mathbf{Qb}\}$$

parameterized by some  $\mathbf{Q} \in \mathbb{R}^{d \times m}$  with  $d < m$ .

First, solve  $\mathbf{QA}\mathbf{x}^0 = \mathbf{Qb}$  for the initial iterate  $\mathbf{x}^0$ . Then, iterate:

$$\mathbf{x}^{k+1} = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{x} - \mathbf{x}^k\|_{\mathbf{B}} \quad \text{such that} \quad \mathbf{S}^k \mathbf{Ax} = \mathbf{S}^k \mathbf{b}, \quad \mathbf{QA}\mathbf{x} = \mathbf{Qb},$$

where  $\|\mathbf{z}\|_{\mathbf{B}} = \sqrt{\mathbf{z}^T \mathbf{B} \mathbf{z}}$  and  $\mathbf{S}^k \sim \mathcal{D}$  are random sketching matrices.

**Upshot:** Same updates as usual sketch-and-project with  $\mathbf{AP}_{\mathbf{B}}$  in place of  $\mathbf{A}$ , where  $\mathbf{P}_{\mathbf{B}}$  is the oblique projector onto  $\text{Null}(\mathbf{QA})$  w.r.t.  $\|\cdot\|_{\mathbf{B}}$ .

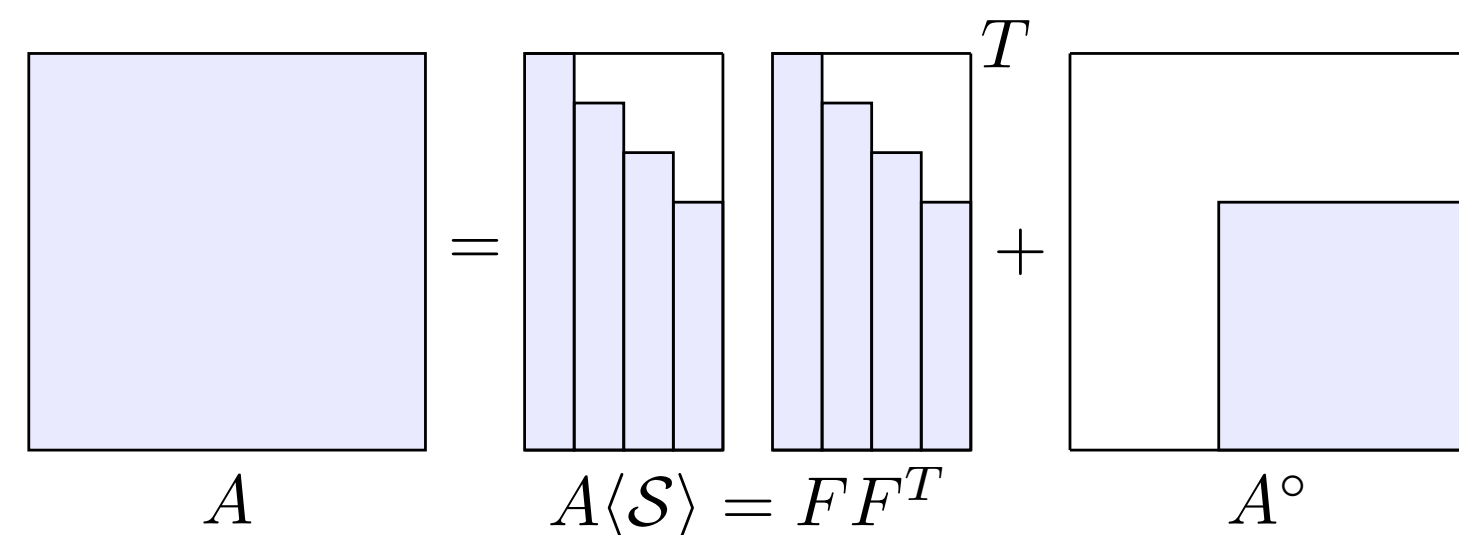
## Solving positive semidefinite linear systems

Specifically, consider solving  $\mathbf{Ax} = \mathbf{b}$ , where  $\mathbf{A} \in \mathbb{R}^{n \times n}$  is psd.

**Example:** *Kernel ridge regression* is equivalent to solving  $(\mathbf{K} + \lambda \mathbf{I})\mathbf{x} = \mathbf{y}$ , where  $\mathbf{K}$  is a kernel matrix measuring the similarity between  $n$  pairs of data points and  $\lambda > 0$  is a regularization parameter.

**Randomized block coordinate descent:** Choose  $\mathbf{B} = \mathbf{A}$  and sparse sketches  $\mathbf{S} = \mathbf{e}_{\mathcal{J}}^T$  corresponding to a block  $\mathcal{J} \subseteq \{1, 2, \dots, n\}$  of size  $\ell$ .

**Subspace constraint:** Choose  $\mathbf{Q} = \mathbf{e}_{\mathcal{S}}^T$  also aligned with coordinate basis. In particular, want to find a set  $\mathcal{S} \subseteq \{1, 2, \dots, n\}$  of  $d$  coordinates corresponding to a *good low-rank Nyström approximation*  $\mathbf{A}\langle \mathcal{S} \rangle$  of  $\mathbf{A}$  such that the *residual matrix*  $\mathbf{A}^\circ = \mathbf{A} - \mathbf{A}\langle \mathcal{S} \rangle \approx 0$ .



A good option is the **RPCholesky** algorithm (Chen et al. [2025]): this computes a near-optimal rank- $d$  psd approximation using  $O(nd)$  memory and  $O(nd^2)$  arithmetic operations via **adaptive diagonal sampling**.

## Subspace-constrained randomized coordinate descent (SC-RCD) method

For solving  $\mathbf{Ax} = \mathbf{b}$  with psd  $\mathbf{A} \in \mathbb{R}^{n \times n}$ :

1. Compute rank- $d$  Nyström approx.  $\mathbf{A}\langle \mathcal{S} \rangle = \mathbf{FF}^T$  (RPCholesky)
2. **Initialize:**  $\mathbf{C} = (\mathbf{F}_{\mathcal{S},:})^{-T} \mathbf{F}^T$ ,  $\mathbf{x}^0 = -\mathbf{e}_{\mathcal{S}}^T (\mathbf{F}_{\mathcal{S},:})^{-T} (\mathbf{F}_{\mathcal{S},:})^{-1} \mathbf{b}$
3. **Iterate:** Sample  $\mathcal{J} = \{j_1, \dots, j_\ell\} \stackrel{\text{i.i.d.}}{\sim} \text{diag}(\mathbf{A}^\circ) / \text{tr}(\mathbf{A}^\circ)$ , compute

$$\boldsymbol{\alpha}^k = (\mathbf{A}_{\mathcal{J},\mathcal{J}} - \mathbf{F}_{\mathcal{J},:} (\mathbf{F}_{\mathcal{J},:})^T)^\dagger ((\mathbf{A}_{:, \mathcal{J}})^T \mathbf{x}^k - \mathbf{b}_{\mathcal{J}}) \in \mathbb{R}^\ell,$$

and update  $d + \ell$  coordinates of  $\mathbf{x}^k$  in  $\mathcal{S} \cup \mathcal{J}$ :

$$\mathbf{x}_{\mathcal{J}}^{k+1} = \mathbf{x}_{\mathcal{J}}^k - \boldsymbol{\alpha}^k, \quad \mathbf{x}_{\mathcal{S}}^{k+1} = \mathbf{x}_{\mathcal{S}}^k + \mathbf{C}_{:, \mathcal{J}} \boldsymbol{\alpha}^k.$$

## Convergence results for SC-RCD

**Theorem 1 (Convergence rate).** Let  $\{\mathbf{x}^k\}_{k \geq 0}$  be the iterates of SC-RCD with subspace constraint  $\mathcal{S}$  corresponding to residual matrix  $\mathbf{A}^\circ = \mathbf{A} - \mathbf{A}\langle \mathcal{S} \rangle$ , and block size  $\ell$  with i.i.d. diagonal sampling. Then, for any solution  $\mathbf{x}^*$ ,

$$\mathbb{E} \|\mathbf{x}^k - \mathbf{x}^*\|_{\mathbf{A}}^2 \leq \left(1 - \frac{\lambda_{\min}^+(\mathbf{A}^\circ)}{\text{tr}(\mathbf{A}^\circ)}\right)^{k\ell} \cdot \|\mathbf{x}^0 - \mathbf{x}^*\|_{\mathbf{A}}^2.$$

**Note:** Iteration complexity improves *at least linearly in the block size  $\ell$*  (more generally, holds for sketch-and-project when  $\mathbf{S}^k$  has i.i.d. rows).

Given an approximation rank  $d$ , let  $r$  be the largest integer satisfying  $d \geq r + r \log(1/\eta_r)$ , where  $\eta_r = \sum_{i>r} \lambda_i(\mathbf{A}) / \sum_{i=1}^n \lambda_i(\mathbf{A})$ . Define the **normalized tail condition number** of  $\mathbf{A}$  by

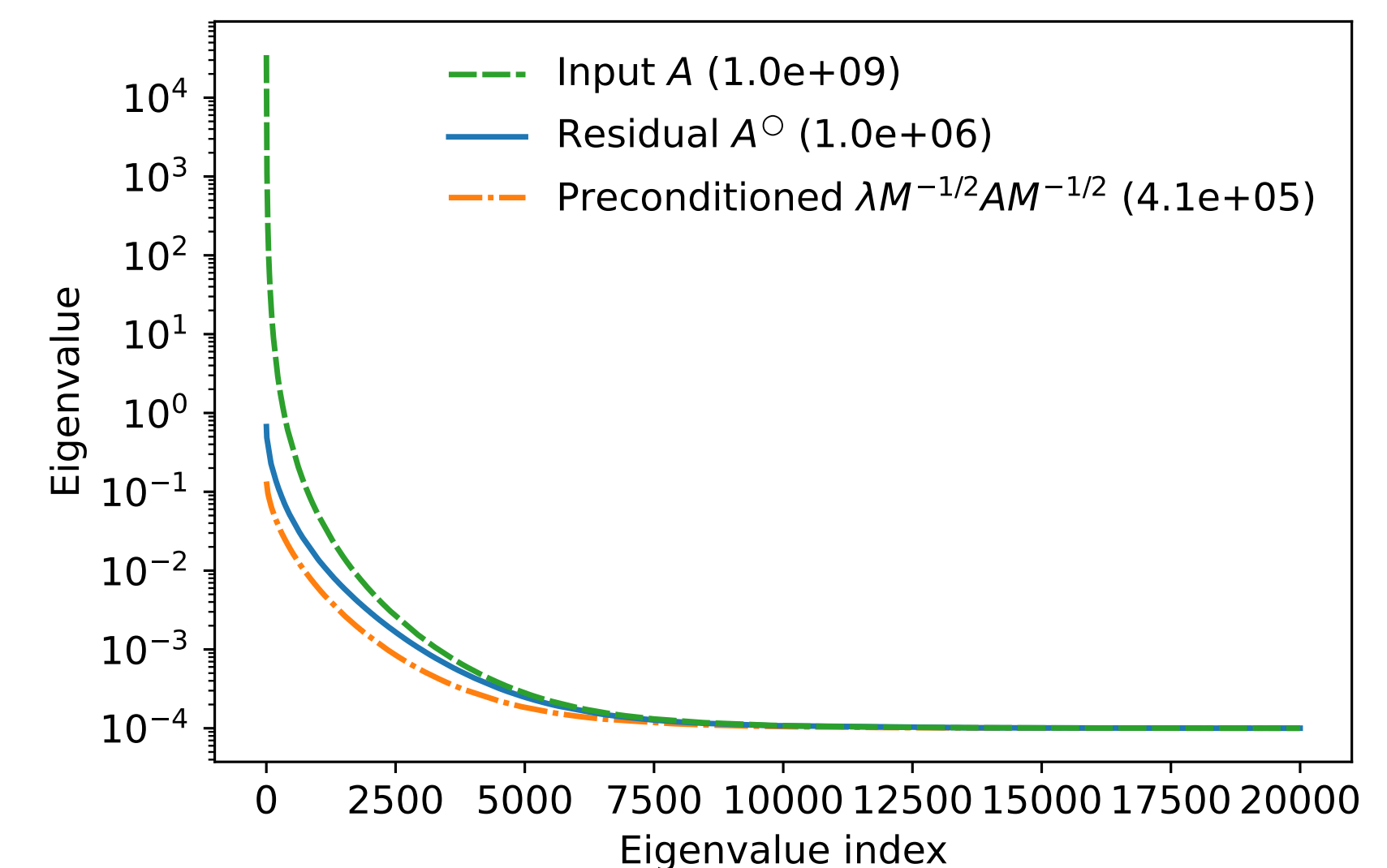
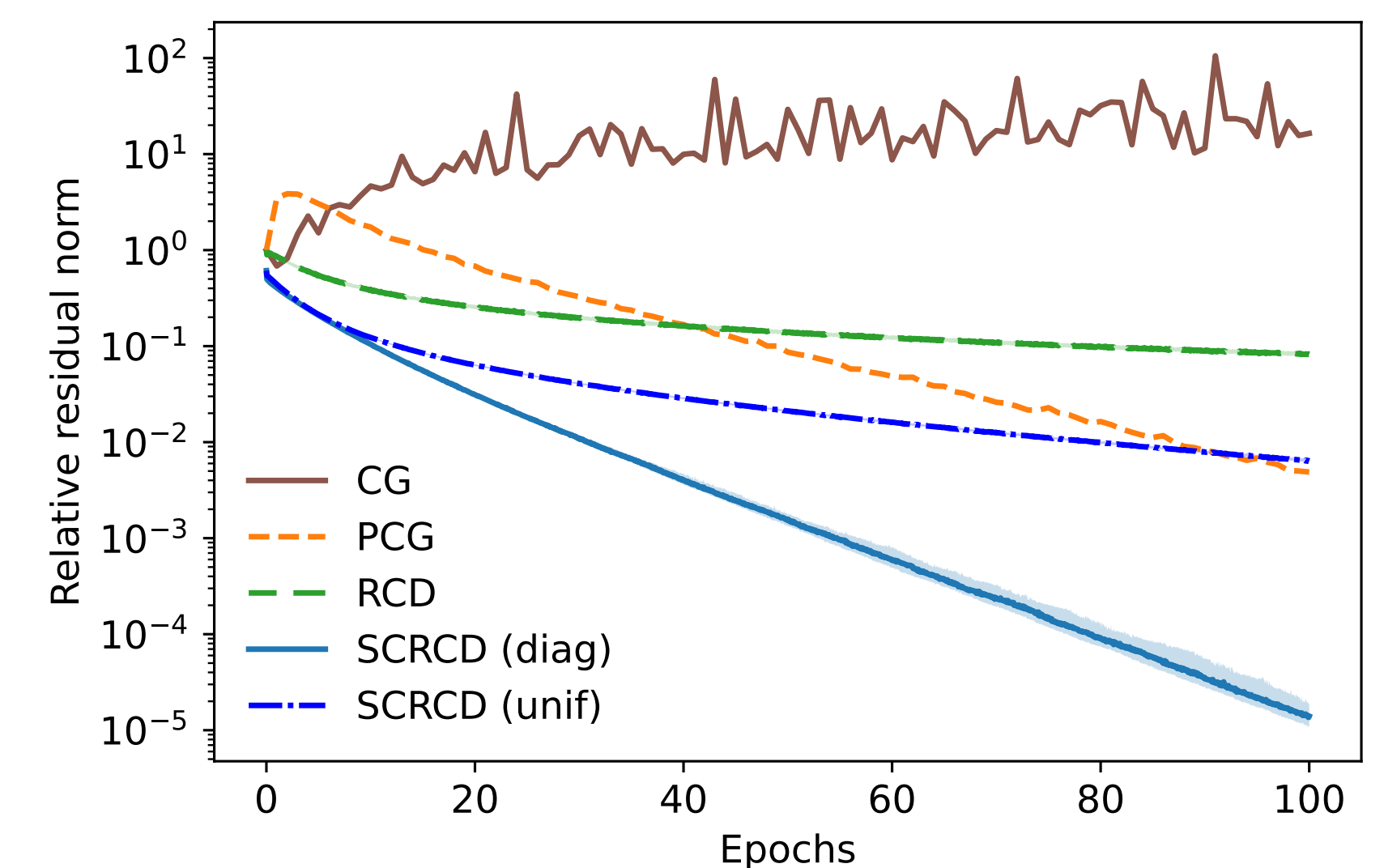
$$\bar{\kappa}_r(\mathbf{A}) := \frac{1}{n-r} \sum_{i>r} \frac{\lambda_i(\mathbf{A})}{\lambda_{\min}^+(\mathbf{A})}.$$

**Theorem 2 (Complexity).** The SC-RCD method with approximation rank  $d$  and block size  $\ell = d$  computes  $\mathbf{x}^k$  satisfying  $\mathbb{E} \|\mathbf{x}^k - \mathbf{x}^*\|_{\mathbf{A}}^2 \leq \varepsilon \|\mathbf{x}^0 - \mathbf{x}^*\|_{\mathbf{A}}^2$  using  $O(nd)$  storage and  $O((n^2 + nd^2) \cdot \bar{\kappa}_r(\mathbf{A}) \log(1/\varepsilon))$  arithmetic operations.

**Note:** Under a mild assumption that  $\mathbf{A}$  is not extremely ill-conditioned,  $r$  can be taken as large as  $r = O(d/\log n)$ . In particular, if  $\bar{\kappa}_r(\mathbf{A}) = O(1)$  for some  $r = O(\sqrt{n}/\log n)$ , then SC-RCD with  $d = \ell = O(\sqrt{n})$  can compute an  $\varepsilon$ -approximate solution using  $O(n^2 \log(1/\varepsilon))$  operations.

## Numerical results

Consider solving a kernel ridge regression problem  $(\mathbf{K} + \lambda \mathbf{I})\mathbf{x} = \mathbf{y}$  with  $n = 100,000$  and tiny regularization  $\lambda = 10^{-9}n$  from a particle physics dataset. (Top) SC-RCD with  $d = \ell = 1,000$  is an effective solver that does not require the kernel matrix  $\mathbf{K}$  to be stored in memory ( $\approx 80\text{GB}$ ). A *preconditioned conjugate gradient method* (PCG) based on RPCholesky (Díaz et al. [2024]) is also included for comparison. (Bottom)  $\mathbf{A} = \mathbf{K} + \lambda \mathbf{I}$  exhibits rapid spectral decay.



## References

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